

Your lateral sensitivity and ISF data folder

A short guide to what is included, where to begin, and what the verification establishes.

The folder contains the empirical lateral sensitivity and intrinsic spatial frequency (ISF) material for **all four thesis subjects**. Subjects **1, 2 and 4** are the main ISF interpretation set. Subject **3** is retained as clearly marked diagnostic data, following the thesis exclusion for fit quality.

What has been prepared

Material	Included
Luminance conditions	17
Original numeric trial files	560
Recorded trials	115,445
Reconstructed staircases	2,469
Saved profile/spread pairs	850,000
Saved fit rows	850,000
Fits passing saved RA FMS screen	280,785

Start here

Open **README.md**, then **docs/MEASUREMENTS.md**. Begin with the staircase thresholds, lateral profiles and ISF endpoint table in **data/processed**. Staircase examples and the ISF error-bar overview are in **figures/measurements**. Your four supplied images are preserved in **figures/reference**.

The original acquisition folders and your separate CSF/PSF repository remain unchanged. No public upload or release has been made for this new folder.

How the data fit together

1. Trial records become sensitivity measurements

Participants judged whether a probe appeared above or below fixation. The staircase changed probe intensity across trials. Reversal points provide contrast-threshold estimates. The base and flanker conditions are compared at corresponding probe positions.

Sensitivity is the reciprocal of the threshold. The stored natural-log ratio compares flanker sensitivity with base sensitivity. A positive value means greater sensitivity with the flanker; a negative value means lower sensitivity.

2. Reversal subsets produce alternative profiles

The deterministic profile uses each staircase's last five detected reversals. The resampling calculation selects five distinct reversals from reversals 2-10, giving 126 possible subsets. At a given position, the same subset indices are used across the included base and flanker staircases.

The archive preserves 50,000 inputs per condition, including deterministic input zero. The resampled spread is recalculated alongside each profile. It describes variability within these data; it is not a population confidence interval.

3. Fitted curves provide ISF estimates

A decaying oscillatory curve is fitted to a spatial log-sensitivity profile. Its frequency gives the ISF estimate in cycles per degree. A frequency-match score (FMS) screens the saved fits. The main summaries use residual-analysis FMS from 0.8 to 1.0 inclusive.

The accepted fitted frequencies form distributions. Candidate clusters, percentile endpoints and display-retention flags are recorded separately. A small cluster does not automatically qualify as a retained mode.

Where to find each stage

data/raw - Preserved numeric trials and de-identified settings.

data/processed - Reversals, lateral profiles, named-column saved fits and mode summaries.

data/bootstrap - All original resampled profiles and paired spreads, in compressed CSV tables.

code - Portable calculations, checks, new seeded resampling and diagnostic plotting.

validation - Detailed integrity, reproduction and comparison results.

A file ending in .gz is compressed. The largest tables are best read with Python rather than loaded into a spreadsheet worksheet. Column definitions are in **data_dictionary.csv**.

What the checks establish

Strong agreement with the archived data

All 17 deterministic profiles and spreads reproduce exactly. The complete resampling audit also passed: all 15,349,693 nonzero-input spatial points agree with a permitted paired reversal-subset calculation to within $1e-12$.

All 17 accepted-fit counts match the thesis. All 13 candidate-cluster count pairs for the primary subjects also match Table 5.2. Cryptographic hashes protect the preserved files and allow every original profile/spread input to be reconstructed from the consolidated tables.

A remaining historical fitting qualification

The supplied current fitting code does not reproduce most of the historical fitted values tested. In a fixed sample, all 85 optimizations completed successfully, but only **12 of 85** matched both saved frequency and FMS to four decimal places. The exact historical writer and its row-to-input mapping have not been established.

The original saved fits and supplied figures are preserved. New representative fits are explicitly labelled as diagnostics. The guide also records a difference between the plotting code's example-selection indexing and choosing the highest saved FMS within a cluster. Neither diagnostic should silently replace a thesis figure.

Other details are kept visible

The documentation records differences between some source settings and thesis labels, between the current peak-finding code and the KDE description, and between the two percentile conventions. Subject 1's archived luminance label 20 versus Table 4.2's value 15.0 is preserved as an unresolved difference. See **docs/KNOWN_LIMITATIONS.md** for the full account.

Before making a public release

Choose the intended licences for data and code, confirm contributor attribution and public-sharing permissions, and decide whether to resolve or retain the documented scientific qualifications. The local archive is prepared for that review. It should have its **own** GitHub/Zenodo release metadata; the existing CSF/PSF DOI does not identify this dataset.

Scientific context: Alexander A. Minns, *Wave-Based Cortical Coupling as a Mechanism for Spatial Frequency Selectivity and Gain Control*, doctoral thesis, Loughborough University, May 2026. Methods and precise thesis locations are listed in **docs/METHODS.md**.